

Project Memo — Arms Supply, Patron Leverage, and Dispute Onset

Status memo, August 11, 2026. Yidong Yang & Jerry Zhang. Supersedes all earlier results documents; see file inventory at the end.

1. One-paragraph summary

The project began as “arms-supplier overlap reduces militarized dispute onset” (MENA pilot, then global). Rigorous diagnostics overturned the original global result — it was an artifact of sample construction. What survived, sharpened, and was then confirmed out-of-sample is a more precise claim: **sharing a dominant arms patron (a single supplier providing $\geq 50\%$ of both sides’ imports) is robustly associated with fewer new militarized disputes — where patron structures are strong: the Cold War era, patron–client (cross-region) dyads, and MENA. Generic portfolio overlap without a dominant patron does nothing, and the effect is absent in the diversified post-1990 market on average and in Asia-Pacific in particular.** The mechanism claim (relational patron leverage: aid, spare parts, conditionality) is supported by timing (dyad fixed effects) but not by a supply-shock IV — so the causal story rests on process evidence, making the Camp David case study load-bearing.

2. How the argument evolved (keep this history — it disciplines the paper)

1. **Original pilot (writing sample):** MENA15, 1990–2014, TERGM; overlap negatively associated with onset.
2. **Global expansion:** full SIPRI bilateral register scraped (all suppliers $\times$ recipients, 1950–2025); MID 5.0, V-Dem, CINC, alliances, contiguity, capital distance (computed), COW trade, UNGA ideal points — all harmonized to COW codes.
3. **First global results (later voided):** overlap OR of about 0.29 with full controls; “identification suite” (UNGA control, dyad FE, non-aligned subsample, Bartik IV) all passed. *These were run on a panel whose inclusion rule (“both import OR MID present”) turned out to select non-importing dyads into the sample only when in conflict.*
4. **Baseline hardening (the turning point):** zero-diagnostic, peace-years controls, standard politically-relevant sample, Firth. On the rebuilt unconditional panel, **generic overlap is null on every margin.** Original global finding: artifact.
5. **Clean MENA re-test:** complete-network design survives directionally; the sharp measure — **shared dominant patron, esp. shared US patron (OR 0.29, $p=0.015$, jackknife-stable)** — is where the action is.
6. **Out-of-sample confirmation (declared primary test):** Cold War 1955–89, global politically-relevant dyads: shared dominant patron **OR 0.594, $p=0.001$** ; shared US patron 0.54 ($p=0.004$); generic overlap also works in the bipolar era (OR 0.49) — consistent with overlap being nearly equivalent to patron-sharing under bipolarity.
7. **Identification, clean panel:** dyad FE passes (full panel OR 0.617, $p=0.002$); **shift-share IV does not confirm** (strong first stage, null reduced form) $\rightarrow$ the market-share component of patron-sharing is not what pacifies; leverage is relational.
8. **Region $\times$ era map:** effect concentrated in cross-region (patron–client) dyads in the Cold War (OR 0.53, $p=0.031$) and directionally in MENA both eras; **Asia-Pacific is a consistent outlier** (null-to-positive).

3. The paper’s thesis as it now stands

Shared dependence on a dominant arms patron constrains the outbreak of militarized disputes between recipients. The constraint operates through the patron’s relational leverage — aid, resupply, spare parts, conditionality — not through portfolio similarity per se. Accordingly it is visible where patrons dominate (Cold War, patron–client dyads, MENA) and vanishes where the arms market is diversified (post-1990 average, Asia-Pacific). Camp David illustrates the mechanism: after 1979 the United States became the dominant patron of both Egypt and Israel, and no Egypt–Israel dispute onset followed.

Contributions: (i) dyadic/network view of arms transfers; (ii) a null that matters (generic overlap does nothing — corrects the intuition the pilot started from); (iii) the patron-leverage conditional finding with out-of-sample confirmation; (iv) mechanism case study; (v) timely implication — as the market multipolarizes (China, second-tier suppliers), this restraint channel erodes.

4. Evidence inventory (what we can claim, at what strength)

claim	evidence	strength
Generic overlap does not pacify (global)	clean panel, all margins, Firth	strong null
Shared dominant patron and fewer onsets, Cold War	declared primary test, $p=0.001$	strong association
Same, MENA post-1990 (shared US)	complete network, $p=0.015$, jackknife-stable	moderate (small n, multiple testing)
Timing consistent within dyads	conditional logit FE, $p=0.002$	moderate-strong
Era/market-structure moderation	CW vs post-90 contrast; interaction $p=0.216$	directional only — do not overclaim
Causal (exogenous) effect	shift-share IV null	not established ; frame as relational-leverage refinement
Asia-Pacific exception	every design/era	consistent pattern, explain in paper

5. Honest limitations to state

Small onset counts in regional cells; shared-patron hypothesis was generated post-hoc in MENA (mitigated by declared out-of-sample confirmation); IV null admits two readings (relational mechanism vs. no causal effect); MID 5.0 ends 2014; alliance/CINC data end 2012 (forward-filled 2013–14).

6. Next steps (priority order)

1. **Camp David case study** (now load-bearing; Poast endorsed). Trace US leverage over Egypt & Israel: aid packages, F-16/spare-part conditionality, crisis episodes. Quantitative backbone from our own data: Egypt’s supplier shift 1970s→80s, Egypt–Israel patron-sharing series, MID history.
2. **Relational leverage measures**: patron share of imports (continuous), dependence asymmetry, aid flows; embargo/cutoff episodes as event-based shocks (better than Bartik here).
3. **Market-concentration moderator**: continuous test of “effect declines as market multipolarizes” (replaces the underpowered era interaction).
4. **Asia-Pacific paragraph**: why leverage fails there (rivalry pairs among same-patron clients).
5. Network robustness (TERGM on clean panels) — appendix.
6. Update Poast; revise the slide deck (current deck reflects the voided results — **do not circulate until revised**).

7. File inventory (build/)

- **Panels:** clean_polrel_panel_1955_2014.csv (primary), clean_mena_panel_1990_2014.csv, clean_polrel_panel_1990_2014.csv; dyad_year_panel_1990_2014.csv (**voided — do not use**).
- **Results memos (current):** RESULTS_clean_reassessment.md, RESULTS_mena_clean.md, RESULTS_patron_confirmation.md, RESULTS_patron_identification.md, RESULTS_region_era.md. (**Voided:** RESULTS_region_logit.md, RESULTS_identification.md, RESULTS_bartik_iv.md.)
- **Scripts:** integrate.py, clean_panel.py, hardening.py, mena_clean.py, patron_confirm.py, patron_identification.py, region_era.py, bartik_iv.py, iv_estimation.py, build_panel.R.
- **Figures:** FIG_patron_by_region_era.png (current), FIG_data_overview.png (descriptive, still valid); FIG_overlap_by_region.png (**voided**).
- **Slides:** overlap_conflict_slides.tex/.pdf (**needs revision before any circulation**).
- **Data acquired (data/):** global SIPRI register, MID 5.0, V-Dem v16, COW Trade 4.0, capital coordinates/distances, Voeten ideal points.